HECTOR GENARO PACHECO

Software Engineer | Embedded Systems • IoT • Physical Systems

Southern California • **hector.genaro.pacheco@gmail.com** • **Portfolio** • **GitHub** • **LinkedIn**

PROFESSIONAL SUMMARY

Software Engineer combining computer science with a professional industrial Instrumentation & Controls background, experienced in both hardware systems engineering and collaborative software development. Track record includes embedded Linux systems on Raspberry Pi 5, native iPadOS applications in Swift/SwiftUI, programmatic camera control via gphoto2 and AVFoundation, real-time sensor telemetry pipelines, Jira/GitLab workflows, and systematic physical-world troubleshooting.

TECHNICAL SKILLS

Languages: Swift, Python, C++, TypeScript, JavaScript, SQL, Bash

Hardware Protocols: AVFoundation, gphoto2, USB PTP, GPS/NMEA (UART/Serial), PIR Motion, 4–20 mA Loops, HART

Networking & APIs: FastAPI, WebSockets, REST APIs, Node.js, Express, Docker

Platforms: Palantir Foundry, Palantir AIP, Human-in-the-Loop AI

Embedded & Linux: Raspberry Pi 5, iPadOS / iOS, Debian/Linux, systemd Services, Kiosk Mode, GPIO, Active Cooling

Industrial Controls: PLC Systems & I/O, Relay Logic, 480VAC/120VAC/24VDC, Fluke 789, Trex HART

Software & Tools: Git, GitLab, Jira, React, PostgreSQL, Prisma ORM, Systematic Fault Isolation, Calibration Logs

Quality & Standards: Signal Validation, Preventative Maintenance, MPE Tolerances, Industrial Auditing

PROFESSIONAL & TECHNICAL EXPERIENCE

Pacific Horizon Labs LLC — *Founder & Software Engineer* COMPANY SITE ↗ PHOTO BOOTH ↗ RISING OPS ↗ **2026 – Present**

- Engineered a commercial event kiosk platform operated under Pacific Horizon Labs LLC, transitioning from an embedded Linux hardware prototype into a native Apple iPadOS SwiftUI application.
- Built foundational generation with active-cooled Raspberry Pi 5, 64-bit Debian, systemd daemons, and gphoto2 USB PTP Canon DSLR automation for unattended event reliability.
- Architected Pacific Rising Ops, an operations automation platform integrating Twilio Voice interactions with deterministic human-in-the-loop approval state machines.

Instrumentation & Controls Technician — *Process Control & Industrial Systems* **Field Engineering Experience**

- Calibrated and commissioned 4–20 mA process transmitters (pressure, temperature, flow, level) and troubleshot PLC I/O wiring, relays, and industrial power loops with systematic fault isolation and auditable logs.

Software Engineering Contributor — *Nonprofit serving underserved communities* **Collaborative Contribution**

- Contributed to an existing codebase within a collaborative team environment, utilizing Jira issue tracking, Git/GitLab workflows, and peer technical communication.

FEATURED SOFTWARE ENGINEERING PROJECTS

FieldTrack AI — *Edge Computing & Sensor Telemetry Platform* (Jul 2026 – Aug 2026) **Raspberry Pi 5 • Python • FastAPI • Node.js • React • WebSockets • GPS/NMEA** GITHUB ↗

- Built an edge telemetry platform on Raspberry Pi 5 integrating GPS/NMEA serial stream parsing, PIR motion hardware interrupts, and system diagnostics.
- Implemented Python/FastAPI hardware services streaming low-latency JSON telemetry frames over bidirectional WebSockets to a real-time React dashboard.

CalTrack — *Industrial Calibration SaaS & Mobile Platform* (Jun 2026 – Jul 2026) **React • TypeScript • Node.js • Express • PostgreSQL • Docker • AWS ECS Fargate** LIVE APP ↗ GITHUB ↗

- Bridged physical industrial instrumentation workflows with modern cloud software, engineering an equipment calibration platform for process plant compliance.
- Modeled database schemas for 4–20 mA transmitters (pressure, temperature, flow, level), tracking calibration histories, tolerances, and calibration cycles.

Additional Repositories & Deployments: **Pi Arcade OS** — Embedded Gaming Platform (Raspberry Pi/Python/GPIO/Pygame) SITE ↗ GITHUB ↗ • **EstateOS** — Agentic Real Estate Platform (Multi-Agent Orchestration & Underwriting) CASE STUDY ↗ • **Procurement Tracking** — Full-Stack GraphQL Workflow Platform LIVE APP ↗ GITHUB ↗

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)
Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026, 16.00 Continued Education Credits)
Palantir Technologies — Foundry & AIP Learning Credentials (May 2026)